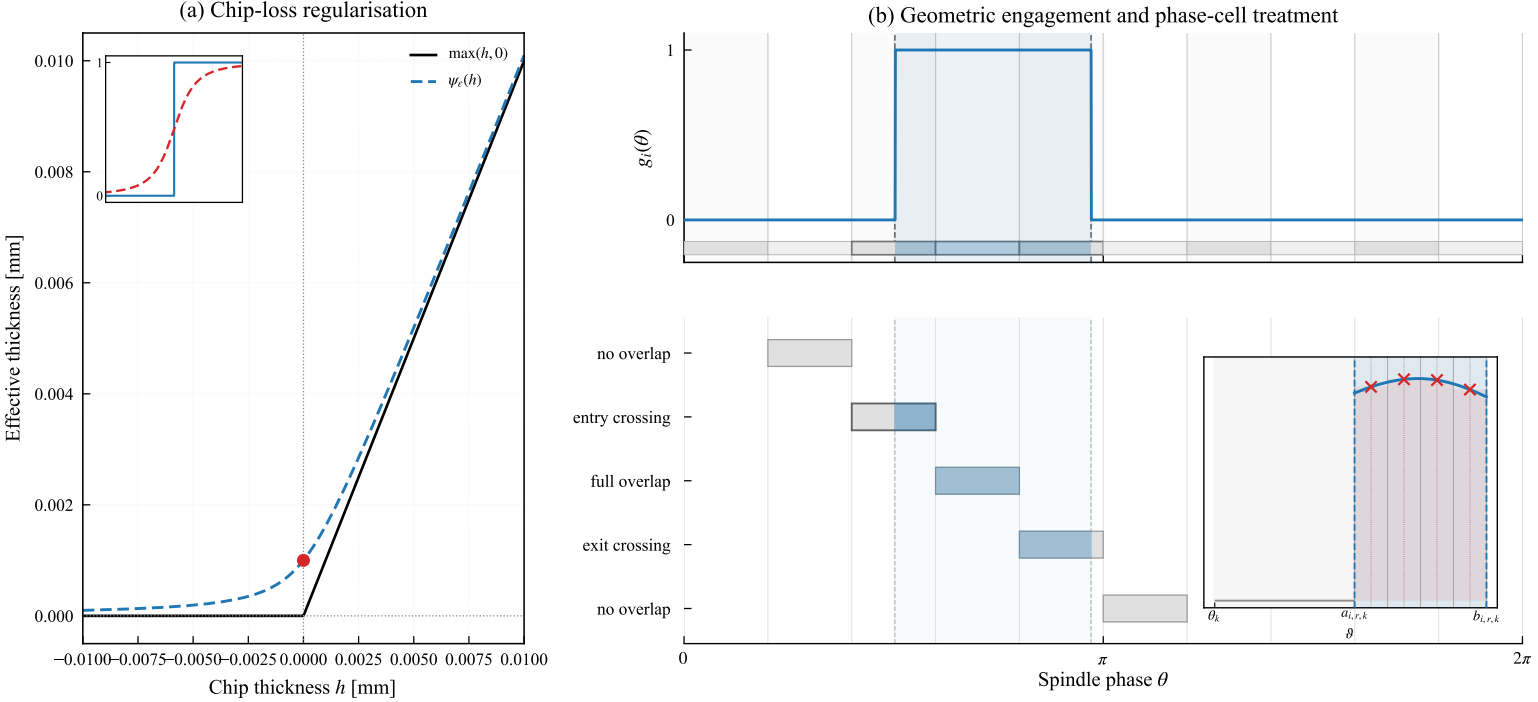
(a) Chip-loss regularisation
max(h, 0)
ψ_ε(h)
Effective thickness [mm]
Chip thickness h [mm]
0.010
0.008
0.006
0.004
0.002
0.000
−0.0100 −0.0075 −0.0050 −0.0025 0.0000 0.0025 0.0050 0.0075 0.0100
1
0
(b) Geometric engagement and phase-cell treatment
g_i(θ)
1
0
no overlap
entry crossing
full overlap
exit crossing
no overlap
θ_k
a_{i,r,k}
b_{i,r,k}
ϑ
0
π
2π
Spindle phase θ